

HYDROLOGICAL CONNECTIVITY IN INTERMITTENT RIVERS AND EPHEMERAL STREAMS

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Andrew J. Boulton^{*}, Robert J. Rolls[†], Kristin L. Jaeger[‡], Thibault Datry[§]

University of New England, Armidale, NSW, Australia^{} University of Canberra, Bruce, ACT, Australia[†] USGS Washington Water Science Center, Tacoma, WA, United States[‡] Irstea, UR MALY, centre de Lyon-Villeurbanne, Villeurbanne, France[§]*

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- Hydrological connectivity extends and interacts over longitudinal, lateral, and vertical dimensions in intermittent rivers and ephemeral streams (IRES), varying more than in perennially flowing systems
- In IRES, hydrological connectivity is primarily controlled by interactions between channel features (e.g., size, shape, substrate composition) and flow regime (e.g., duration and timing of streamflow), often modified by human activities
- Flow intermittence disrupts hydrological connectivity at multiple scales, creating nonlinear and patchy “discontinua” that strongly influence physical, chemical, and ecological patterns and processes in space and time
- A proposed conceptual framework explores three aspects of hydrological connectivity and potential effects of intermittence

2.3.1 INTRODUCTION

In all streams and rivers, hydrological connectivity, defined as the “water-mediated transfer of matter, energy, or organisms within and/or between elements of the hydrologic cycle” (Pringle, 2003: 2685), has three spatial dimensions—longitudinal, lateral, and vertical—that interact along a fourth dimension, time (Ward, 1989). In intermittent rivers and ephemeral streams (hereafter, IRES), cessation of surface flow disrupts hydrological connectivity in one or more spatial dimensions, with repercussions for most physical, chemical, and biological processes (Fig. 2.3.1). Longitudinally, cessation of surface flow halts downstream transport of sediments, other materials, and biota (Hooke, 2003; Rolls et al., 2012), and usually heralds the onset of drying of shallow channel sections, especially riffles. Laterally, aquatic habitats on the floodplain and along the riparian zone that were hydrologically linked to the main channel during overbank flows or through bank storage become isolated when water levels decline (Fig. 2.3.1), interrupting the two-way transfer of energy, sediments, and various organisms (Nakano and Murakami, 2001; Paetzold et al., 2006). Vertically, most of the exchange of water between the surface channel and the shallow saturated sediments below (i.e., hyporheic zone, White, 1993) ceases when surface flow stops, impairing processes such as oxygenation of the hyporheic zone by

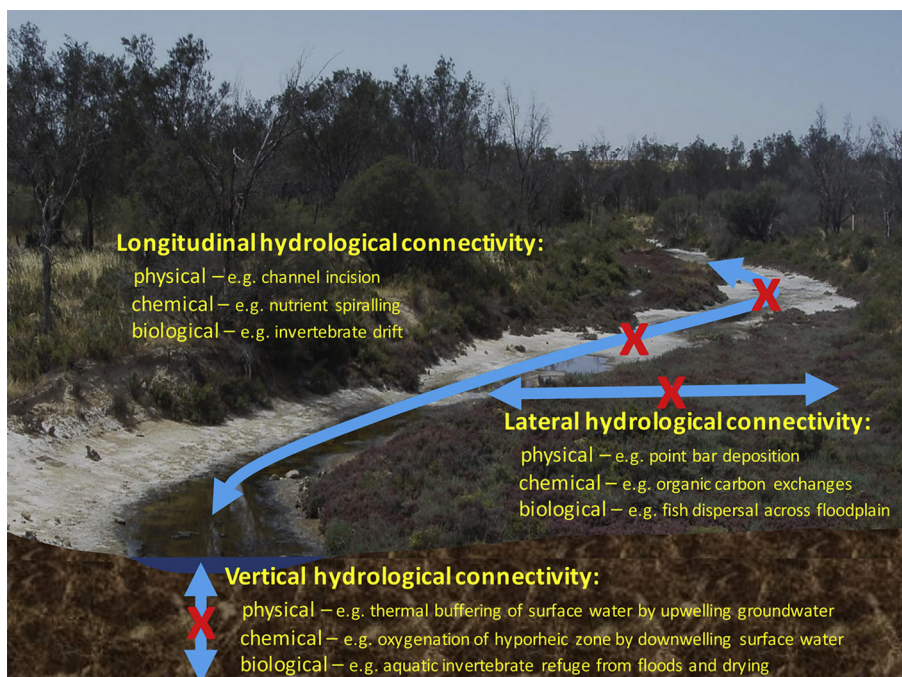


FIG. 2.3.1

Flow cessation and drying in IRES interrupt physical, chemical, and biological processes that rely on hydrological connectivity along three spatial dimensions—longitudinal, lateral, and vertical—portrayed as *blue lines*. The *double-headed arrows* imply many processes operate in both directions, including downstream to upstream (e.g., fish migration). Interruption of hydrological connectivity is indicated by *red crosses*.

Photo: Courtesy A. Boulton.

downwelling water or the flux of nutrients to the surface water in upwelling areas (Datry and Larned, 2008; Boulton et al., 2010, Fig. 2.3.1).

Thus, flow and hydrological connectivity are tightly linked, often creating complex hydrological patterns along all three spatial dimensions. In this chapter, “flow” refers to the net movement of water in a riverscape; where necessary, surface flow is explicitly distinguished from subsurface flow because IRES are defined by intermittence of surface flow even though flow may continue in the sediments below and alongside the channel. In IRES, hydrological connectivity in all three spatial dimensions is disrupted by intermittence interacting with geomorphology (Chapter 2.1) and flow regime (Chapter 2.2).

In IRES, hydrological connectivity spans the full spatial spectrum from the entire stream system (*sensu* Frissell et al., 1986, $>10^3$ m), exemplified by longitudinal connectivity from headwaters to the river terminus, to within microhabitat systems ($<10^{-1}$ m, Frissell et al., 1986) where hydrological connectivity along a few millimeters of interstitial flowpaths can generate sharply contrasting conditions of redox potential (Hendricks and White, 1991). Although hydrological connectivity also occurs in perennially flowing streams, the key difference in IRES is that streamflow intermittence dramatically affects hydrological connectivity in all three dimensions and promotes its dynamic nature in space and time. This spatiotemporal dynamism in hydrological connectivity exerts powerful influences on the

biogeochemistry, biota, and ecosystem processes in all IRES, providing unique research opportunities to exploit such variability in connectivity to advance our understanding of general stream ecology. (e.g., [Datry et al., 2014](#); [Jaeger et al., 2014](#); chapters in Sections 3 and 4).

In this chapter, we start by reviewing the factors driving hydrological connectivity at multiple spatial scales and the implications of these drivers for different IRES. We then describe patterns of hydrological connectivity occurring along each of the three spatial dimensions and how they are affected by geomorphological features ([Chapter 2.1](#)) and flow regime ([Chapter 2.2](#)), especially intermittence. A preliminary conceptual framework, drawing on ecological connectivity theory ([Beger et al., 2010](#)), is proposed as a basis for generating hypotheses that predict physical, chemical, and biological responses to intermittence along interacting hydrological dimensions. We conclude by synthesizing the key aspects of hydrological connectivity in IRES by way of an introduction to the next three sections of this book.

2.3.2 WHAT GOVERNS HYDROLOGICAL CONNECTIVITY IN IRES?

Hydrological connectivity along longitudinal, lateral, and vertical dimensions in IRES is largely governed by interactions between the multiple components of flow regime (magnitude, frequency, duration, timing, and rate of change of flow, [Chapter 2.2](#)) and fluvial geomorphology (e.g., channel and floodplain shape, size, gradient, sediment composition, and location along the network). Many of these components are modified by catchment land use and other human activities ([Fig. 2.3.2](#)). Other important drivers of hydrological connectivity at a landscape scale include climatic conditions such as timing and amounts of precipitation and evaporation, hydrogeological features governing gains and losses of groundwater along the channel, tectonic activity (e.g., fault lines, volcanism), and underlying lithology and geology, including geological transitions along the channel ([Fig. 2.3.2](#); [Graf, 1988](#); [Tooth and Nanson, 2011](#)). At finer scales, patterns of hydrological connectivity in IRES may be influenced by biological processes ([Fig. 2.3.2](#)) such as microbial activity and benthic algae altering substrate permeability ([Mendoza-Lera and Mutz, 2013](#); [Hartwig and Borchardt, 2015](#)) or the presence of vegetation within the channel modifying surface flows and bed topography ([Sandercock et al., 2007](#); [Sandercock and Hooke, 2011](#)).

In IRES, hydrological connectivity in all three dimensions can change markedly over time. Natural streams and rivers are spatially dynamic ecosystems that undergo cycles of expansion and contraction of their aquatic boundaries in response to variations in flow regime ([Allan and Castillo, 2007](#); [Larned et al., 2011](#)). However, as IRES have especially variable flow regimes ([Chapter 2.2](#)), these cycles of expansion and contraction often include fragmentation of surface water hydrological connectivity during drying ([Stanley et al., 1997](#); [Jaeger and Olden, 2012](#)). The fluctuations of expansion, contraction, and fragmentation generate dynamic mosaics of habitat patches ([Larned et al., 2010](#); [Datry et al., 2014](#); [Chapter 4.9](#)) with their own environmental conditions, biota, and ecological functions.

Along all three spatial dimensions, the changing cycles of expansion, contraction, and fragmentation in IRES produce a hydrological “discontinuum” (*sensu* [Poole, 2002](#)). Thus, hydrological patterns along these dimensions are seldom gradual trends over distance but instead exist as sequences or networks of diverse patches with varying streamflow, presence of water, and hydrological connectivity, especially in highly intermittent IRES. This is consistent with the patch dynamics perspective ([Pickett and White, 1985](#)) that has proved successful for describing ecohydrological patterns at multiple scales in river ecosystems ([Fausch et al., 2002](#); [Winemiller et al., 2010](#)). These ecohydrological patterns are controlled by the pervasive gravitational movement of water downstream and vertically which imposes

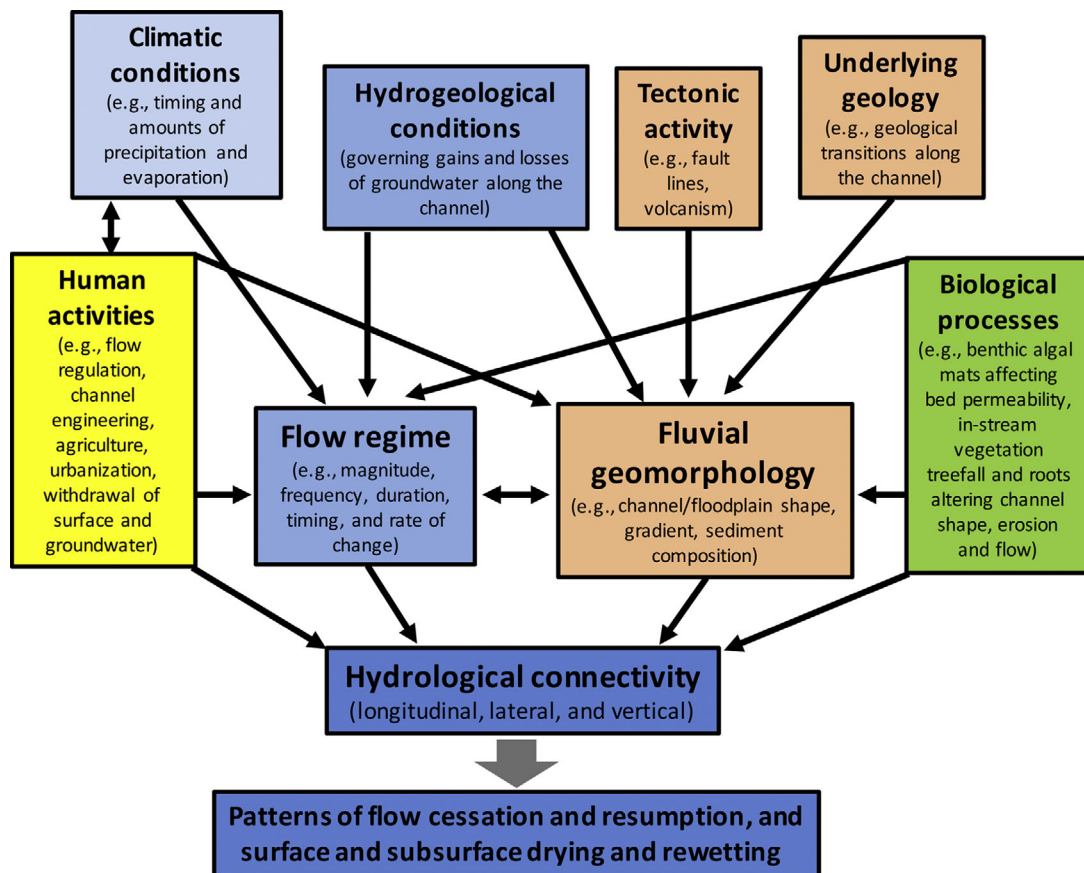


FIG. 2.3.2

Interactions among factors affecting hydrological connectivity in longitudinal, lateral, and vertical dimensions in IRES.

a “catchment hierarchy” (Townsend, 1996), acknowledging that environmental features at a given location along the river network typically reflect both local and upstream hydrological conditions.

Location along the river network also governs the importance of other drivers of hydrological connectivity at that point (Fig. 2.3.2). For example, in the upper reaches of many IRES, precipitation is a strong driver of in-channel flow but this coupling weakens in lowland reaches where groundwater inputs increase (Tooth and Nanson, 2011). Where headwaters of IRES arise in hills and mountains, channel shape is largely bedrock-controlled and strongly influenced by local geology compared with downstream reaches where alluvial sediments and groundwater exchanges are more prevalent (Chapter 2.1; Stanford and Ward, 1993; Tonina and Buffington, 2009). However, all portions of the river network are subject to various human activities that affect hydrological connectivity along longitudinal, lateral, and vertical dimensions (Fig. 2.3.2), either directly such as through water extraction or indirectly from altered flow regimes and/or channel and floodplain geomorphology (Costigan et al., 2016).

These activities include modifications to topography, vegetation cover, runoff, and water quality through agriculture, mining and urbanization, changes to channels by in-stream works (e.g., weirs, dams), and withdrawals of surface water and groundwater (Chapter 5.1).

2.3.3 HYDROLOGICAL CONNECTIVITY, INTERMITTENCE, AND SURFACE WATER DRYING AND REWETTING IN IRES

The characteristic hydrological feature of all IRES is intermittence (Chapter 1), often followed by drying of part or all of the channel (Fig. 2.3.3). Rewetting usually coincides with the onset of sustained flow although in some IRES, there may be several “false starts” where surface water is present or pools

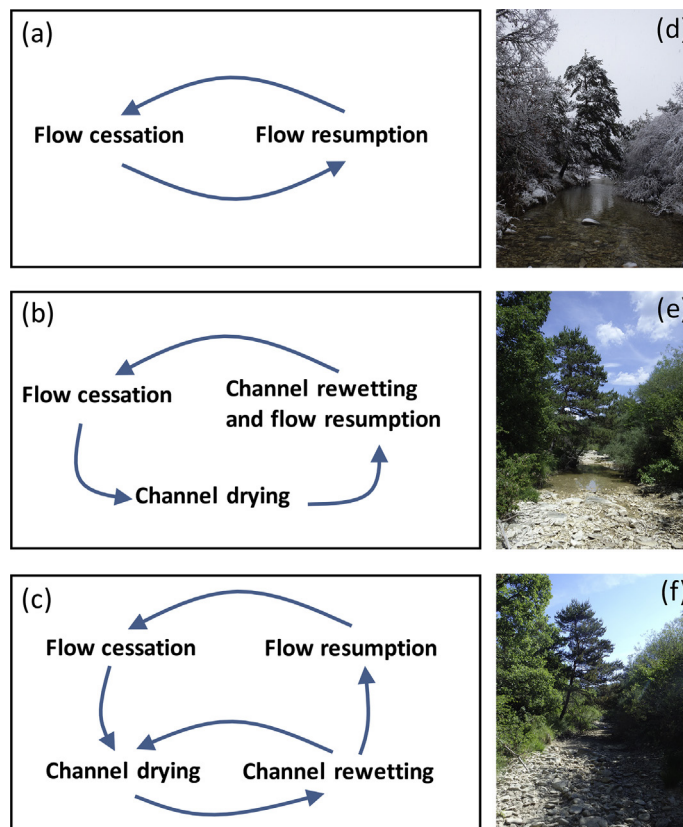


FIG. 2.3.3

Three common sequences of hydrological transitions governing hydrological connectivity in IRES. Surface-flow cessation either involves no channel drying (a), channel drying followed by rewetting and immediate flow resumption (b), or several “false starts” of drying and rewetting before surface flow resumes in earnest (c). Snapshots of this sequence in a French intermittent stream illustrate the flowing (d), drying (e), and dry (f) stages.

Photos: Courtesy B. Launay.

fill and then dry for several weeks or even months before streamflow begins in earnest (e.g., Boulton and Lake, 1990; Bhamjee et al., 2016). Drying in most IRES is a prolonged and gradual “ramp” disturbance (Lake, 2003) whereas rewetting is usually faster, especially when it coincides with resumption of surface flow (Fig. 2.3.3). This hysteresis in channel wetting and drying occurs in nearly all IRES, even those in temperate regions (e.g., Day, 1983).

Another difference between patterns of drying and rewetting is their directionality along the river channel. In most IRES, rewetting extends in a downstream direction, either as surface runoff down the channel (Fisher et al., 1982; Jacobson et al., 2000) or inputs from rejected recharge springs and seeps (Shaw, 2016) arising from unconfined aquifers that initiate expression of surface water into the channel (Doering et al., 2007; Larned et al., 2011), subject to regional geology and geomorphology (Section 2.3.4, Eltahir and Yeh, 1999). In contrast, drying typically occurs in both upstream and downstream directions, particularly in “losing” streams where the water table lies below the channel (Mansell and Hussey, 2005). Again, this pattern of drying is controlled by regional geomorphology (Larned et al., 2011) and groundwater levels relative to the stream channel (Shaw, 2016) as well as exposure to solar radiation and evaporation.

Rewetting and drying patterns in IRES occur across multiple spatial extents (Stanley et al., 1997; Figure 4 in Costigan et al., 2016). At the scale of tributaries and individual river segments, localized precipitation can lead to rewetting and surface flow resumption in some channels whereas adjacent ones stay dry (e.g., Figure 2 in Datry et al., 2014). This asynchrony in rewetting can be highly unpredictable in arid and semiarid regions where brief storms cause flash floods to surge down channels in ephemeral streams that dry within days or weeks (Fig. 2.3.4). In these areas, surface runoff is rapidly initiated, Hortonian overland flow is the dominant source of water, and



FIG. 2.3.4

Flash flood down the channel of an intermittent stream in the Flinders Ranges, South Australia. This channel was completely dry only three hours before the photograph was taken, and the flood originated from a brief but heavy rainstorm in the headwaters.

Photo: Courtesy A. Boulton.

the hydrographs are sharply peaked (Tooth and Nanson, 2011). Further, such spatial variability can vary among events, seasons, and years (Costigan et al., 2015), contributing to geomorphological discontinua (Section 2.3.4) and habitat heterogeneity in IRES at broad scales. There are also broad-scale differences in patterns and rates of drying, reflecting spatial differences in transmission losses along channels (e.g., seepage, evaporation) and local climatic conditions. Human activities extracting surface water, groundwater, or both also influence drying patterns; in some parts of the world (e.g., Australia, Reinfelds et al., 2004), extraction from IRES is governed by regulations such as cease-to-pump rules to minimize ecological harm from rapid anthropogenic water drawdown or extended periods of intermittence.

At stream-reach and finer spatial extents, patterns of drying and rewetting are often asynchronous within IRES, largely governed by channel topography because depressions fill earlier and hold water longer than surrounding areas (Woodward et al., 2015). These fine-scale patterns of drying and wetting create heterogeneous mosaics of habitat patches of varying permanence, with implications for physicochemical (Chapters 3.1 and 3.2) and ecological processes (Chapters 4.7–4.9). This fine-scale heterogeneity also has an important temporal component governed by flow regime operating at the broader scales of the stream segment or river network. For example, rewetting in IRES may coincide with bed-moving floods that erode new pool basins in the channel while filling old ones and depositing sediments across bars and floodplains (Hooke, 2016). These floods flush mats of desiccated algae and leaf litter downstream (Corti and Datry, 2012), redistributing habitats while resetting succession trajectories for benthic algae, aquatic invertebrates, and other organisms (Fisher et al., 1982; Rosado et al., 2015). Again, we see evidence of the catchment hierarchy (Townsend, 1996) but with fine-scale patterns of drying and rewetting nested within broader-scale patterns of surface flow resumption and cessation in IRES.

Ultimately, patterns and variability in hydrological connectivity in IRES arise from spatial and temporal differences in the water budget. Put simply, flow and presence of surface water are a function of water inputs (e.g., precipitation, groundwater), storage (within, below, and alongside channels and floodplains), and outputs (e.g., seepage, evaporation) at different scales (Gordon et al., 2004; Godsey and Kirchner, 2014). In all IRES, across spatial extents from the entire river network to the smallest microhabitat, drying occurs when water losses exceed gains whereas rewetting occurs when gains exceed losses of water. Different patterns of drying and rewetting at multiple spatial scales in IRES create constantly changing hydrological connectivity in all three dimensions, modified by multiple drivers that include channel geomorphology and location within the riverscape.

2.3.4 LONGITUDINAL HYDROLOGICAL CONNECTIVITY IN IRES

In all rivers, gravity drives unidirectional flow of water down the physical gradients of channels. This typically results in longitudinal trends in geomorphology (e.g., channel width, median substrate particle size), hydrology (e.g., median current velocity, discharge), water chemistry (e.g., dissolved oxygen, total dissolved solids), and, ultimately, ecology (e.g., distribution of aquatic plants and animals) (Allan and Castillo, 2007; Boulton et al., 2014).

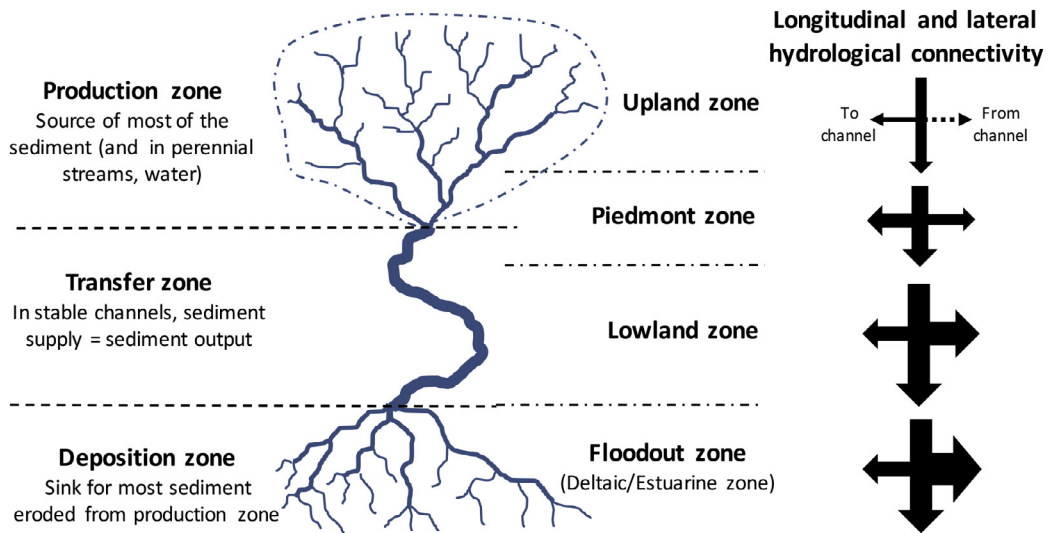
Broadly speaking, longitudinal geomorphological processes in IRES resemble those typical of perennial fluvial systems although there are some distinct differences as described in Chapter 2.1. In an idealized perennial fluvial system (Schumm, 1977), sediment production predominates in the upland

zone, sediment transport in the transfer zone, and sediment deposition in the lower floodout zone which either enters an inland water body, peters out in uncoordinated channels, or exits into an estuary or directly to the coast. The trend of surface expression of longitudinal hydrological connectivity in an idealized perennial system is for progressively larger volumes of water down the network in response to increasing drainage area and channel capacity. Except during overbank events, longitudinal hydrological dimensions of surface flow and volume are constrained by channel capacity and configuration (Gordon et al., 2004). However, with changes in slope, geology, and geomorphology, differences in channel form in the different zones also influence the relative amounts of water exchanging laterally onto floodplains (Fig. 2.3.5A) and vertically into the sediments below and beside the channel (Fig. 2.3.5B, Stanford and Ward, 1993).

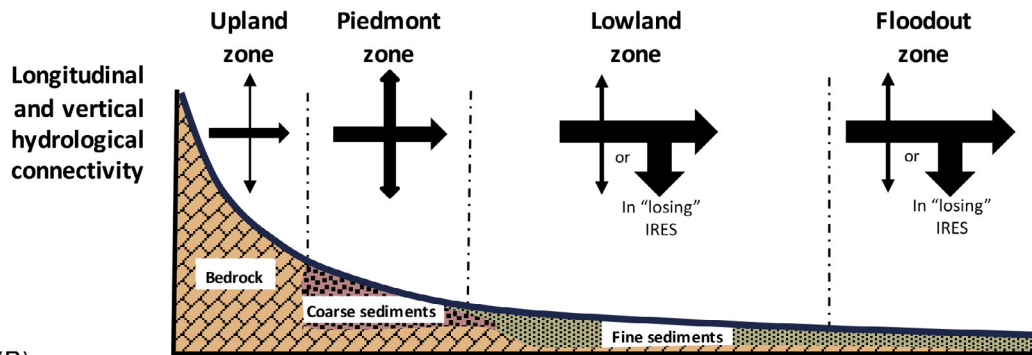
At any point in time and space, channel morphology reflects the interaction of hydraulic forces, substrate resistance, and sediment supply (Schumm, 1977). In many IRES, these interactions are discontinuous because most of the channel adjustment and sediment transport occurs during brief floods followed by long periods of very low or zero surface flow (Wohl, 2014). Crossing thresholds associated with flood magnitude or sediment accumulation leads to sharp changes in channel behavior (e.g., Patton and Schumm, 1981; Gómez-Gutiérrez et al., 2009). These geomorphological discontinua and thresholds mean that longitudinal hydrological connectivity and flow intermittence in IRES are not necessarily a predictable function of channel profile, topography, and bed composition (Tooth and Nanson, 2000; Wohl, 2014). Instead, flow cessation, drying, and rewetting may occur along different sections of the channel according to other drivers such as local patterns of alluvial porosity and permeability, precipitation and evaporation, and groundwater gains and losses (Larned et al., 2011; Costigan et al., 2016). This diversity of drivers, acting in various combinations across the broad geographic spread of IRES globally (Chapter 1), creates an equally diverse array of spatial patterns of intermittence, drying, and rewetting.

To help classify this diversity at the network scale, Lake (2003) described three broad patterns of drying along streams affected by drought (Box 2.3.1). To these three patterns, Datry (2016) added two more: complete drying and point drying. Complete drying is when entire river networks remain dry for weeks or months, only flowing briefly before drying again within days or even hours. This pattern is common in ephemeral streams that lie above the water table, exemplified by many streams in arid (Stanley et al., 1997) and karstic (Meyer and Meyer, 2000) environments as well as in temperate IRES with fluctuating groundwater tables (Winter et al., 1998). Point drying is also common (Godsey and Kirchner, 2014; Datry et al., 2016) and is when multiple isolated sections along the channel dry, interspersed by variable lengths of perennial surface water (Chapter 4.9).

Variations in longitudinal hydrological connectivity generated by these patterns of drying in IRES arise from the interplay between discharge generated by different water gains (e.g., precipitation, groundwater) and transmission losses (Fig. 2.3.6). Transmission losses arise from (i) infiltration and recharge to channel and/or floodplain sediments; (ii) evapotranspiration; and (iii) ponding in terminal storages, such as ephemeral pools, channels, and other wetlands (Knighton and Nanson, 1994). The relative magnitudes of these three losses are governed by catchment area, channel gradient, flow regime, and climatic aridity as well as the geomorphological variables discussed earlier. For example, in many IRES draining small- to mid-sized catchments with low or moderate channel gradients and coarse sediments, infiltration and recharge to shallow aquifers is the major transmission loss (e.g., Hughes and Sami, 1992; Constantz et al., 2002). In contrast, the Orange River, South Africa, largely underlain by bedrock, flows through an extremely hot region and evapotranspiration is the major transmission loss (McKenzie and Craig, 2001).



(A)



(B)

FIG. 2.3.5

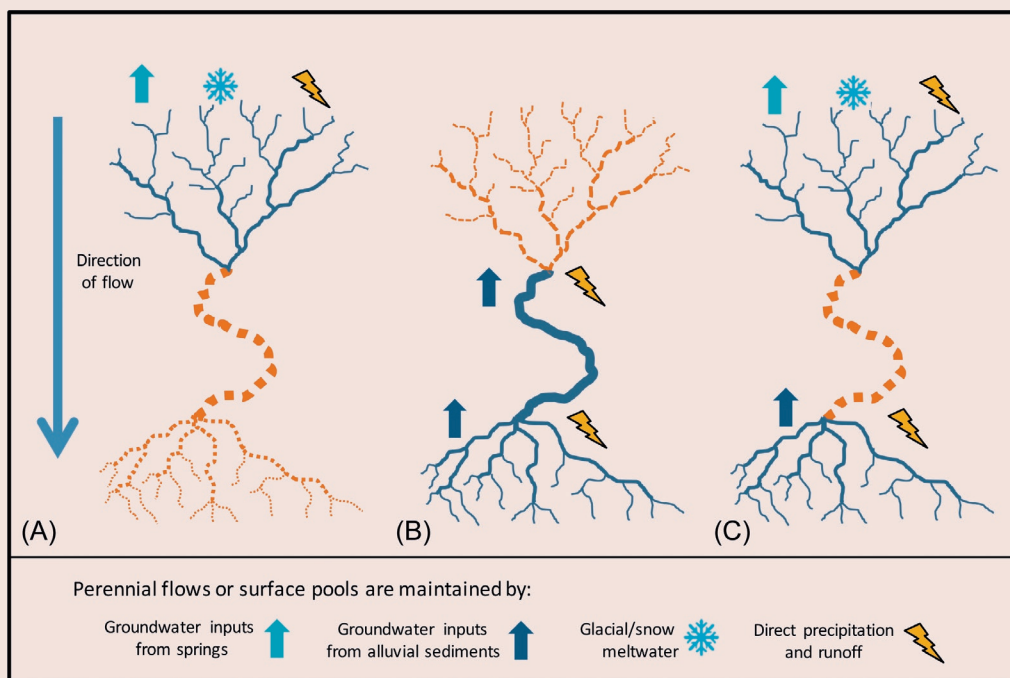
Plan (A) and transverse (B) views of geomorphological zones and their relative longitudinal and lateral hydrological connectivity (*black arrows*; thickness represents relative magnitudes of flow; *broken line* indicates minimal flow) in an idealized single-thread river network. In lowland and floodout zones of "losing" IRES (Fig. 2.3.10), transmission losses of surface water (e.g., infiltration and recharge) are often greater, represented by the large down-pointing *arrows* to the right.

The converse to drying is rewetting. Patterns of rewetting are seldom simply a reversal of drying patterns (Fig. 2.3.3) because the causes of the two events differ. Longitudinally, rewetting in many IRES is triggered by increased precipitation and runoff in the headwaters, sometimes intense enough to cause flash floods that swiftly transform downstream channels from completely dry to flowing strongly (Jacobson et al., 2000; Cohen and Laronne, 2005). Alternatively, the increase in catchment precipitation and runoff can be slower, recharging alluvial aquifers and causing more gradual rewetting

BOX 2.3.1 THREE COMMON PATTERNS OF LONGITUDINAL DRYING

At least three broad longitudinal patterns of drying occur in perennial streams during drought (Lake, 2003) and also apply to many IRES. The first is where streams arise from permanent sources (e.g., springs, headwater lakes) and the upper reaches retain perennial flow whereas the mid and lower reaches dry either completely or to isolated pools (Fig. 2.3.6A). This form is common in IRES whose headwaters arise in mountain ranges or well-watered uplands before draining into lowland channels that extend out into more arid areas (Boulton and Williams, 1996; Jaeger and Olden, 2012). Other situations where this pattern arises are when the river course reaches a highly permeable alluvial plain where infiltration losses exceed inflows (Datry, 2012) and in many karstic systems (Meyer and Meyer, 2000).

The second is where the headwaters dry out but flow persists in the lower reaches (Fig. 2.3.6B). In most of these cases, surface water in piedmont and lowland zones is sustained by groundwater inputs, facilitated by larger and deeper channels downstream (Lake, 2003). Datry et al. (2016) describe IRES in Bolivia where a fall in groundwater causes headwater drying yet lowland reaches remain perennial owing to their deeper and less permeable channels. The third common pattern is when the mid-reaches dry while the headwaters and lower reaches retain flow or permanent water (Fig. 2.3.6C). In the spatially intermittent Selwyn River, New Zealand, the upper reaches sustain perennial flow for about three kilometers and although “losing”, are supplied by hillslope runoff which peters out before a 45-km segment in the mid-reaches with highly variable rates of drying (Arscott et al., 2010). The next 8 km of the channel intersect confined alluvial aquifers at or near the surface, the river becomes “gaining,” and perennial flow is reestablished, this time primarily fed by groundwater instead of runoff. Other examples of this mid-reach drying pattern include IRES in northern Victoria, Australia (Lake, 2003) and Sycamore Creek in Arizona, USA (Stanley et al., 1997).

**FIG. 2.3.6**

Three broad patterns of drying (*dotted brown lines*) occur in IRES and drought-affected perennial streams: (A) drying in mid- and lower reaches, (B) drying in upper reaches, and (C) drying in mid-reaches.



FIG. 2.3.7

Floodwaters spread from the channel of Narran River, Australia, (in 2008) across vast inland floodplains, triggering “boom” periods of biological productivity.

Photo: Courtesy P. Terrill.

driven by a combination of runoff, streamflow, and groundwater inputs (Stanley et al., 1997; Tockner et al., 2000). Initial floods often suspend large amounts of organic material that have accumulated in the channel during the dry phase and carry them downstream, with implications for downstream water quality (e.g., pulses of anoxic water, Hladyz et al., 2011; Chapter 3.2).

Most rewetting events in inland Australian IRES with vast floodplains ($>10,000\text{km}^2$) are relatively short-lived and transmission losses are high (e.g., 70%–98% in the Diamantina River, Costelloe et al., 2003). However, occasionally heavy rain will initiate huge slow-moving floods that take many weeks or even months to reach terminal wetlands such as Lake Eyre, hundreds of kilometers downstream. As these floods move down along the channel, they spread laterally across immense floodplains (Fig. 2.3.7) whose clays swell and restrict infiltration once their crack capacity is exceeded (Knighton and Nanson, 1994). These longitudinal and lateral rewetting events trigger extraordinary “booms” in ecological productivity eventually followed by “bust” periods when drying occurs (Bunn et al., 2006; Leigh et al., 2010).

2.3.5 LATERAL HYDROLOGICAL CONNECTIVITY IN IRES

At the river network scale, the size and predominant directionality of the lateral hydrological dimensions in IRES, as in perennial rivers, are primarily a function of location along the river network (Fig. 2.3.5), correlated with differences in channel topography, form, and substrate composition among the main geomorphological zones (Box 2.3.2). The generalities that follow acknowledge that substantial finer-scale variation exists along IRES, especially in how channel and floodplain morphology are altered by geology, vegetation patterns, and human activities, and that this longitudinal variation inevitably affects patterns of lateral hydrological connectivity.

Dimensions of lateral hydrological connectivity in upland zones are typically shorter than in piedmont, lowland, and floodout zones because of the differences in channel cross-section, gradient, and extent of floodplain development (Box 2.3.2). Floodplains in lowland and floodout zones are usually well developed (Fig. 2.3.8C) and may be so vast that during extreme floods, lateral hydrological connectivity of surface water extends far from the main channel or channels (Fig. 2.3.7), transporting sediments, organic matter, and aquatic biota for long distances (Bunn et al., 2006; Muehlbauer et al., 2014). Further, in contrast to upland and piedmont zones, the directionality of hydrological connectivity in lowland and floodout zones can be complex, resulting in variable rates and locations of the delivery of water and materials to adjacent riparian and other terrestrial ecosystems. Surface water floods in lowland and floodout zones of IRES have attracted much attention from ecologists interested in how organisms exploit transient lateral hydrological connectivity to colonize new habitats (e.g., fish, Kerezszy et al., 2013; Chapter 4.5), emerge from resting stages in sediments (e.g., zooplankton, Jenkins and Boulton, 2007; Chapter 4.8), and redistribute energy and organic material in complex aquatic-terrestrial subsidies (Bunn et al., 2003; Chapter 4.7).

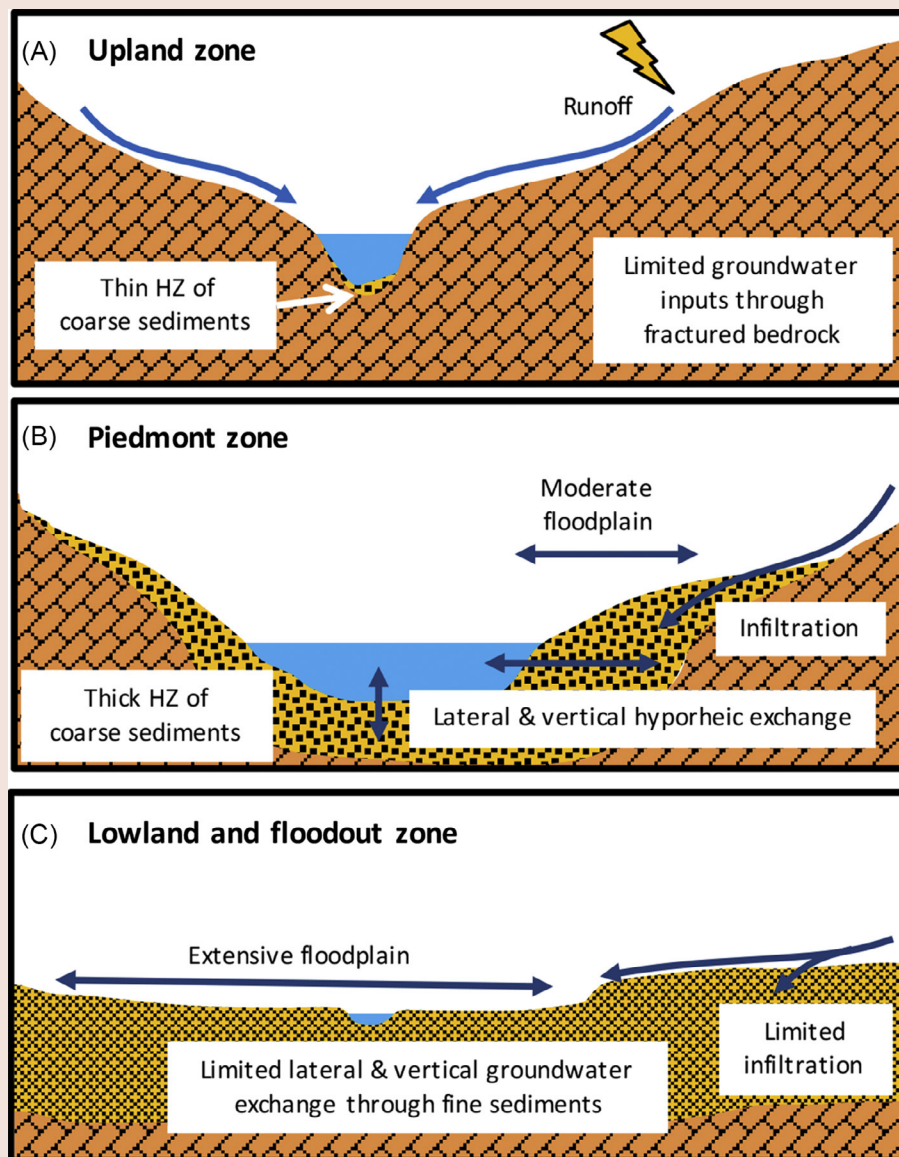
The spatial extent, flow rates, and residence time of surface water spreading laterally across the floodplains of IRES are strongly controlled by the topography, substrate composition, and vegetation cover of the floodplain because these govern runoff and rates of evapotranspiration and infiltration (Hamilton, 2010). Although lateral hydrological connectivity in many IRES is complex, there are several broad trends associated with the different zones along the river network. One trend is that the extent and residence time of surface floodwaters in piedmont zones are typically shorter than on the floodplains of lowland and floodout zones. This is because in piedmont zones, compared to lowland and floodout zones, travel distances of water from the channel onto the floodplain are often shorter

BOX 2.3.2 CHANNEL FORM AND SUBSTRATE COMPOSITION IN GEOMORPHOLOGICAL ZONES ALONG IRES

Channels in the upland zone are usually narrow V- or U-shaped single-thread or sometimes multi-threaded channels, often overlying bedrock, with a low width-to-depth ratio, steep banks, and limited if any floodplain (Fig. 2.3.8A; Sutfin et al., 2014). Consequently, lateral hydrological connections in the upland zones of IRES are short because of the relatively short travel distances down hillslopes to the channel. Surface flow along these connections is usually toward the channel (Fig. 2.3.5A), and rewetting along this dimension occurs in response to precipitation, overland flow, and subsurface interflow (Lake et al., 2006). During overbank flows, these lateral inputs are briefly reversed away from the channel (Rassam et al., 2006).

In the piedmont zone, channels start to broaden, increase in width-to-depth ratio, and become more rectangular in cross-section while the banks are less steep and may include narrow floodplains (Fig. 2.3.8B, Singer and Michaelides, 2014). Channels are single or multi-thread but can vary widely (Sutfin et al., 2014) and gradients are usually less steep than in the upland zone. Bed sediments typically comprise poorly sorted mixtures of coarse sediments (e.g., cobbles, pebbles) with large voids among the particles. Lateral hydrological connections frequently include subsurface flowpaths in this zone (Fig. 2.3.5) with much faster transmission rates than in lowland and floodout zones where the sediments are finer.

In the lowland and floodout zones, single-thread channels are very broad with high width-to-depth ratios and may be “under-fit” with surface flow usually occupying only a small part of the channel (Fig. 2.3.8C). Commonly, these two low-gradient zones are drained by multi-thread braided and anastomosing channels, varying in stability and geomorphic complexity (Chapter 2.1). Bed sediments are typically fine, dominated by sands and clays although sometimes larger particles and even outcropping bedrock may occur (Tooth and Nanson, 2000).

BOX 2.3.2 CHANNEL FORM AND SUBSTRATE COMPOSITION IN GEOMORPHOLOGICAL ZONES ALONG IRES (Cont'd)

FIG. 2.3.8

In many IRES, the typical channel cross-section form, extent of floodplain development, and substrate composition vary among zones: (A) in upland zones, steep-banked V- or U-shaped channels overlie bedrock and have minimal floodplain development; (B) in piedmont zones, shallow-banked rectangle-shaped channels overlie coarse, poorly sorted alluvium and have moderate floodplain development; (C) in lowland and floodout zones, very shallow-banked broad channels overlie fine, well-sorted alluvium and have extensive floodplains. Only single-thread channels are shown here but there may be multiple channels (see text). HZ, hyporheic zone. Geomorphological zones illustrated in [Fig. 2.3.5](#).

(Box 2.3.2), evapotranspiration demands are lower, and lateral and vertical infiltration rates are greater because of this zone's coarser sediments (Tonina and Buffington, 2009; Fig. 2.3.8).

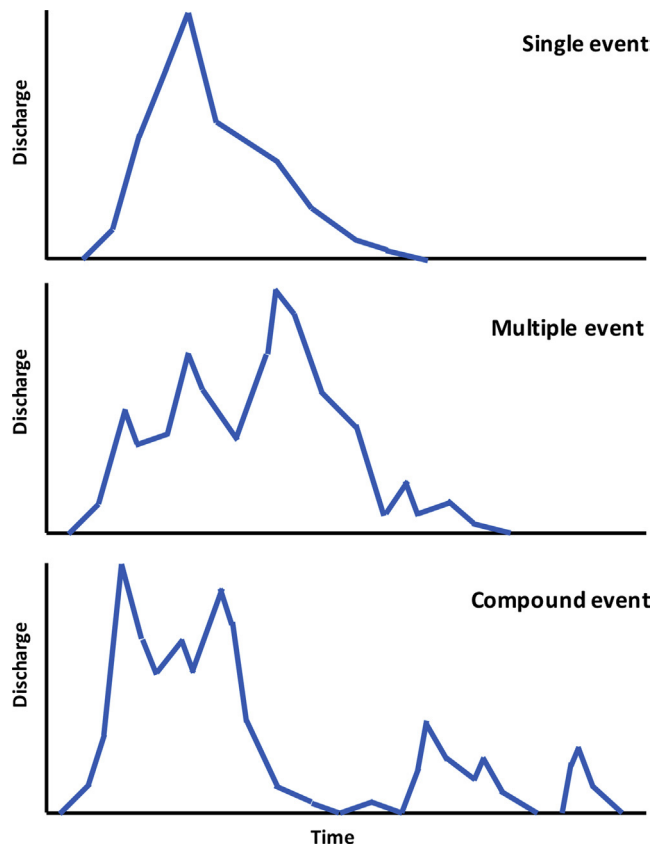
Another trend is that when infiltration capacity is exceeded, flow velocities of surface water on floodplains of piedmont zones are generally faster than on the floodplains of lowland and floodout zones. These faster flows are caused by the steeper slopes; in lowland and floodout zones, floodplain topographic gradients may be extremely low (e.g., 0.0005–0.002, Tooth, 2005) and floodwaters move more sluggishly, further slowed by the drag resistance of inundated vegetation. Implications of this difference among geomorphological zones in flood velocities include contrasting rates of lateral erosion and deposition (Chapter 2.1), altered water residence times, and different hydraulic conditions for aquatic biota. Other IRES with well-developed floodplains are braided rivers, and again, major differences in their lateral hydrological connectivity arise from the interplay of gradient, substrate porosity, and groundwater table. This interplay creates complex spatial patterns where almost-perennial channels adjoin highly intermittent ones (e.g., Selwyn River, Arscott et al., 2010). During floods, water extends laterally across these braids, temporarily reducing spatial variability in water quality across different channels (Thomaz et al., 2007).

Finally, lateral dimensions of hydrological connectivity arising from overbank flows in IRES are strongly controlled by the type of flood and the shape of its hydrograph, associated with location along the river network. Flash floods, as their name implies, are short-lived events (Fig. 2.3.9) that travel swiftly as roiling pulses of water and suspended material that can substantially modify channel form and bed composition in many IRES (Cohen and Laronne, 2005), causing dramatic but only transient increases in lateral connectivity. Flash floods in IRES are more typical of upland and piedmont zones than lowland and floodout zones where rainfall runoff in the catchment travels longer distances on average (dissipating the water's energy) and floodwaters are able to spread laterally in the broader channels and across the wider floodplains downstream.

However, not all floods in IRES are simple one-off events, prompting Knighton and Nanson (2001) to classify floods as single, multiple, or compound according to the number of flood peaks and the shapes of the rising and falling limbs of the flood hydrograph (Fig. 2.3.9; Table 2.3.1). These different types of floods have different implications for longitudinal, lateral, and vertical hydrological connectivity (Table 2.3.1), especially the frequency and duration of intermittence. For example, although a single-peak flood may initiate surface flow in a previously dry channel at a given location along a river, the duration of subsequent flow may be shorter than after a multiple or compound flood with the same peak discharge as the single-peak event (Leigh et al., 2010). Further, the timing and sequence of individual flood events as well as the combination of different types will have variable effects on patterns of intermittence and, consequently, longitudinal, lateral, and vertical hydrological connectivity at multiple scales. Flood timing and sequence affect ecological processes and biota in many IRES (Arscott et al., 2010; Leigh and Datry, 2017).

2.3.6 VERTICAL HYDROLOGICAL CONNECTIVITY IN IRES

Apart from evaporation and direct rainfall, vertical hydrological connectivity in IRES arises from water exchanging between the surface (channel and inundated floodplain) and the subsurface. This hydrological dimension governs surface-flow intermittence in piedmont, lowland, and floodout zones of IRES where groundwater gains and losses dominate the water budget (Fig. 2.3.5). Conversely, in streams with fine-grained sediments, infiltrated water is stored in the vadose zone and depleted by evapotranspiration. Intermittence in streams with limited vertical hydrological connectivity is largely

**FIG. 2.3.9**

Hydrographs of three different types of flood events classified by [Knighton and Nanson \(2001\)](#) according to the number of flood peaks and the shapes of the rising and falling limbs. Implications of these flood events for longitudinal, lateral, and vertical hydrological connectivity are presented in [Table 2.3.1](#).

determined by the spatial variability of rainfall-runoff than by groundwater inputs, and spatial patchiness of rainfall may even lead to independent streamflow events in different reaches of the same network, sometimes never connecting.

When streamflow is lost from a channel to groundwater or the subsurface, the stream is described as “losing” or influent ([Fig. 2.3.10A](#)) whereas groundwater enters the channel in “gaining” or effluent streams ([Winter et al., 1998](#); [Fig. 2.3.10B](#)). Along its length, a channel can alternate between gaining and losing in response to temporal fluctuations in water table elevation as well as longitudinal variations in geology, topography, substrate composition, and other factors ([Cardenas, 2009](#)). Therefore, at a given location in a river network, fluctuations in the elevation of the water table (e.g., in response to precipitation or evapotranspiration) can reverse the direction of vertical flow. The same can happen when high surface flows pass down a river ([Malard et al., 2002](#)), causing a transient change in bank storage that recharges the channel after the flows have passed.

Two key points must be made. The first is that the spatial patterns and cycles of surface flow cessation, drying, rewetting, and flow resumption largely result from the dynamics of gaining and losing exchanges in many IRES as a function of the water table level relative to the channel. Although surface runoff usually triggers rewetting and streamflow in upland zones, rising groundwater levels in response to recharge by precipitation often initiate rewetting and surface flow resumption in piedmont and possibly lowland zones of IRES as their channels become gaining, especially in IRES with shallow water tables or in extremely permeable catchments (e.g., karsts, Meyer and Meyer, 2000). When water tables fall below the channel (Fig. 2.3.10A), surface flows and water levels decline; prolonged losing conditions can result in cessation of surface flow and, ultimately, surface and subsurface drying (Boulton, 2003; Larned et al., 2011). So pivotal to flow intermittence is this alternation of gaining and losing conditions, it is used by many hydrologists (Gordon et al., 2004) to distinguish ephemeral streams as

Table 2.3.1 Floods in IRES can be classified according to the number of flood peaks and the shapes of the rising and falling limbs of the flood hydrograph (Fig. 2.3.9; Knighton and Nanson, 2001), and these have implications for hydrological connectivity in longitudinal, lateral, and vertical dimensions

Event and characteristic features (Knighton and Nanson, 2001)	Implications for each hydrological dimension (examples)		
	Longitudinal	Lateral	Vertical
<i>Single event</i>			
One well-defined peak discharge to which a progressive rise from zero is followed by a progressive fall to zero (Fig. 2.3.9A). Often arising from isolated rainfall event as flash flood	Rewet channel sediments and fill channel pools may initiate brief periods of flow. As flash floods may erode sediments and transport leaf litter down channel, often as a pulse of material	Unless an especially large event, only limited effects on surface lateral hydrological connectivity for a short distance onto the floodplain and water does not persist. Very limited recharge through floodplain sediments	In piedmont zones, rewet channel hyporheic sediments reestablish subsurface flowpaths. After bed-moving flash floods, hydraulic conductivity may be high in reworked sediments and new gravel bars
<i>Multiple event</i>			
More than one peak discharge, with a progressive increase in the magnitude of peaks on the rising limb toward a well-defined maximum, followed by a progressive decrease in the magnitude of peaks on the falling limb as flow returns to zero (Fig. 2.3.9B). Often arising from sequences of rainfall events or combined effect of lagged tributary inflows	Rewet channel sediments and fill channel pools initiate sustained flow. Progressive increases in peaks may erode sediments in different parts of channel, promoting in-stream geomorphological complexity. Particulate organic matter (e.g., leaf litter accumulated during dry periods in the channel and on the banks) is carried downstream in multiple pulses, declining after the maximum peak	Progressively greater areas of riparian zone and floodplain in lowland and floodout zones are inundated laterally as peaks approach the maximum. If subsequent peaks occur soon after the maximum, their additive effect further lengthens lateral hydrological connectivity. Pulsed extensions to lateral hydrological dimensions have geomorphological and ecological implications	Hyporheic flowpaths are reestablished in piedmont zones; longer flows promote complex biogeochemical gradients along subsurface flowpaths. Pulsed events alter advective patterns and water quality of downwelling water. Infiltration and recharge of floodplain sediments are substantial and may subsequently return water to the channel to prolong postflood flow

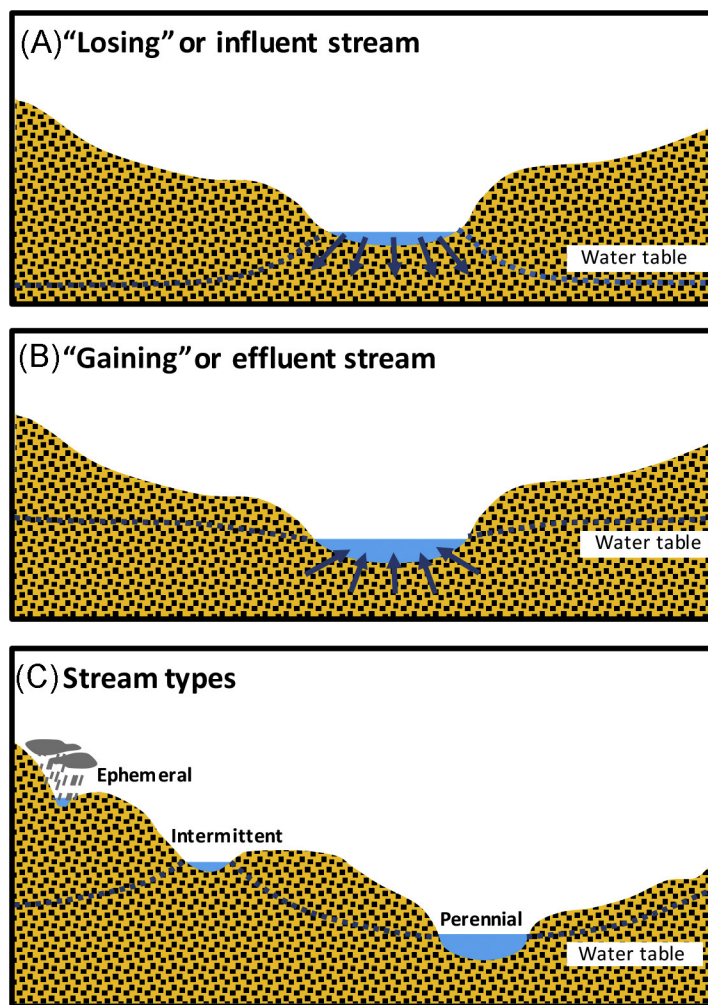
Table 2.3.1 Floods in IRES can be classified according to the number of flood peaks and the shapes of the rising and falling limbs of the flood hydrograph (Fig. 2.3.9; Knighton and Nanson, 2001), and these have implications for hydrological connectivity in longitudinal, lateral, and vertical dimensions—cont'd

Event and characteristic features (Knighton and Nanson, 2001)	Implications for each hydrological dimension (examples)		
	Longitudinal	Lateral	Vertical
<i>Compound event</i>			
More than one peak discharge, with nonprogressive rises and falls before or after the maximum discharge (Fig. 2.3.9C). Often arising from sequences of rainfall events or combined effect of lagged tributary inflows	As in multiple events, rewet channel sediments and fill channel pools initiate sustained flow periods. However, pulsed scour and transport of sediments and particulate organic matter are more idiosyncratic, potentially promoting greater microhabitat diversity than for multiple events	Unlike multiple events, inundation of riparian zone and floodplain in lowland and floodout zones may not increase progressively in spatial extent, depending on the shape of the hydrograph. Different lateral hydrological extensions may arise after each event. These extensions are still pulsed but less predictably	As in multiple events, hyporheic flowpaths and associated complex biogeochemical gradients are reestablished in piedmont zones because of longer flows. Pulsed events alter advective patterns and water quality of downwelling water but less predictably. Substantial infiltration and recharge of floodplain sediments may subsequently return water to the channel to prolong postflood flows

channels always lying above the water table and constantly losing, intermittent rivers that fluctuate between gaining and losing, and perennial systems that are predominantly gaining (Fig. 2.3.10C).

The second key point is that the direction and water source of pathways of vertical hydrological connectivity are largely dictated by whether conditions are gaining or losing. Trends in physical and chemical conditions along a vertical flowpath in a losing section of a stream reach where the source water is exposed to the surface differ greatly from those along a vertical flowpath below a gaining reach where the source water is groundwater. For example, along a vertical flowpath of water downwelling from the surface into the hyporheic zone, concentrations of dissolved oxygen decline while concentrations of reduced forms of nutrients usually increase (Stanley and Boulton, 1995; Capderrey et al., 2013). Where nutrient-enriched alluvial groundwater upwells back into the surface stream, there may be local patches of enhanced primary productivity where benthic algae flourish, benefiting from the increased concentrations of nutrients (Fig. 2.3.11).

Much of our understanding of the functional significance to surface stream ecosystems of hyporheic biogeochemical processes and vertical hydrological connectivity arose from pioneering work by Grimm and Fisher (1984) in Sycamore Creek, an Arizonan IRES. Between floods, the surface stream is shallow (<20 cm) and meanders down a broad channel (Fisher et al., 2004), with most of the stream's discharge occurring along subsurface flowpaths in the alluvial gravels below the stream or in the adjacent parafluvial zone. Along these subsurface flowpaths occur sequences of biogeochemical processes (Fig. 2.3.12A) governed by factors such as redox potential, availability and lability of organic matter and rates of microbial activity (e.g., Holmes et al., 1994; Jones et al., 1995).

**FIG. 2.3.10**

"Losing" or influent streams lose groundwater from their channel (A) whereas groundwater enters the channel in "gaining" or effluent streams (B). In a landscape context (C), ephemeral streams are viewed by many hydrologists as channels always lying above the water table and constantly losing, intermittent streams as fluctuating between gaining and losing, and perennial streams as predominantly gaining.

As a result of this sequence of biogeochemical processes, the chemical composition of upwelling water is largely determined by the length of the subsurface flowpath. For example, water upwelling at the end of long subsurface flowpaths, such as those that might occur between widely isolated pools during drying (Fig. 2.3.12B), would be expected to be rich in reduced forms of nitrogen whereas at the end of short subsurface flowpaths (Fig. 2.3.12C), more oxidized forms such as nitrate would dominate.

**FIG. 2.3.11**

Upwelling nutrient-rich water at the end of a subsurface flowpath promotes algal growth (mainly filamentous *Spirogyra* spp.) in intermittent Brachina Creek, Flinders Ranges, South Australia.

Photo: Courtesy A. Boulton.

The length of subsurface flowpaths, the dominant biogeochemical processes, and the water chemistry of upwelling water all vary temporally in Sycamore Creek in response to floods, surface flow cessation, and drying (Jones et al., 1995; Fisher et al., 2004). There is also substantial fine-scale patchiness and variations in solute-specific processing lengths along the multiple flowpaths, adding to the biogeochemical heterogeneity in these systems (Lewis et al., 2007).

Since these seminal studies in Sycamore Creek, similar trends in concentrations of nutrients and other materials along vertical flowpaths have been identified in IRES elsewhere (Boulton and Williams, 1996; Gómez et al., 2009). Likewise, these hydrological trends have been shown to be strongly associated with spatial trends in microbial activity and invertebrate distribution in various IRES (e.g., Claret and Boulton, 2003; Capderrey et al., 2013; Larned and Datry, 2013). Although much of the empirical work on vertical hydrological connectivity in IRES has been done at the reach scale, the additive and interactive effects of subsurface flowpaths and their collective influence on physicochemical heterogeneity at broad scales have also attracted attention, with particular focus on the influence of flooding and intermittence on the spatial configuration and lengths of subsurface and surface flowpaths (e.g., Fisher et al., 1998; Malard et al., 2002). For example, the combined effects of upwelling water from a sequence of several short flowpaths are likely to generate physicochemical conditions very different from those downstream of a single long flowpath (Malard et al., 2002; Capderrey et al., 2013). Network-scale spatial configurations of hydrological connectivity in all three dimensions constantly vary in many IRES owing to the prevailing cycles of surface flow intermittence.

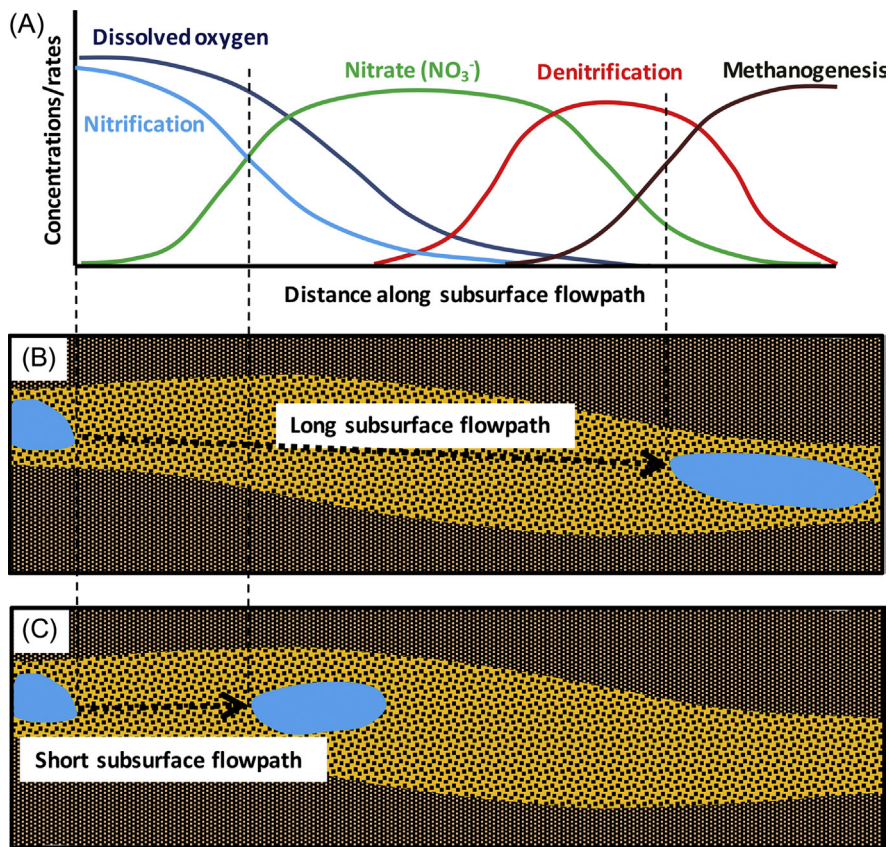


FIG. 2.3.12

Along subsurface flowpaths occur sequences of biogeochemical processes, presented as hypothetical trends (A). Consequently, water chemistry of upwelling water at the ends of a long flowpath (plan-view, B) is likely to differ from that at the end of a short flowpath (plan-view, C), illustrated by the vertical *broken lines* between the top panel and lower two panels.

Trends in (A) are redrawn from Malard, F., Tockner, K., Dole-Olivier, M.-J., Ward, J.V., 2002. A landscape perspective of surface-subsurface hydrological exchanges in river corridors. Freshw. Biol. 47, 621–640, Figure 4, with permission.

2.3.7 A PRELIMINARY CONCEPTUAL FRAMEWORK FOR EXPLORING INTERMITTENCE, CONNECTIVITY, AND INTERACTING HYDROLOGICAL DIMENSIONS IN IRES

Inevitably, disruption of connectivity along one hydrological dimension during intermittence has repercussions for hydrological patterns along the other two dimensions, creating a constantly changing mosaic of interacting flowpaths with different directions and lengths in three dimensions (Poole et al., 2006, 2008). Currently, these three-dimensional hydrological interactions in IRES pose substantial knowledge gaps, ranging from technical ones (Jaeger and Olden, 2012; Bhamjee et al., 2016) through

to conceptual gaps such as which aspects of these hydrological dimensions are the most relevant for predicting biogeochemical and ecological responses to different patterns of intermittence in IRES.

To help address this conceptual knowledge gap, we introduce a preliminary framework that draws on theory about the types of interfaces and ecological connectivity. Our logic in adopting this perspective is that many of the implications of intermittence in IRES relate directly to water's role as an agent of connectivity (Sponseller et al., 2013). Examples range from nutrient spiraling and organic matter transport (Chapter 3.2) to the transfer of genetic material among metapopulations (Chapter 4.9). Our framework aims to help researchers refine their research questions about specific aspects of hydrological connectivity when exploring biogeochemical and ecological effects of intermittence in IRES. For example, one hydrological aspect seldom explicitly stated in discussions of the effects of intermittence is the distinction between the presence of water and the presence of flow yet this distinction is crucial because it dictates whether advection is likely to play a role along a given hydrological dimension. In IRES, the situation is further complicated because at times, flowing and nonflowing conditions cooccur.

Thus, the first criterion of the framework (Table 2.3.2) is whether the hydrological connectivity is mediated by flowing water, and therefore influenced by features associated with spatial and temporal patterns of the flow regime (Poff et al., 1997; Chapter 2.2). In the absence of flow, hydrological connectivity in all three dimensions relies on the presence of surface water, subsurface water, or both. However, lacking transport by flowing water, gradients in concentrations of dissolved and suspended

Table 2.3.2 Preliminary conceptual framework for hypothesizing effects of intermittence on hydrological connectivity and geomorphological, biogeochemical, and ecological processes in different geomorphological zones in IRES

Criterion	Hypothesized differences in processes and effects of intermittence in different geomorphological zones in IRES (examples)		
	Upland	Piedmont	Lowland/Floodout
<i>1. Hydrological connectivity mediated by flow?</i>			
Yes	Promotes erosion of channel and transport of sediments and organic matter	Underpins advective exchange of nutrients between surface and hyporheic water	Carries fine sediments and organic matter from upstream out to floodplain and downstream
No	Increases likelihood that drying will fragment longitudinal hydrological connectivity into isolated pools with implications for water quality	Loss of advective exchange but infiltration of surface water may still recharge alluvial aquifers	Allows silt to settle on inundated floodplain which may reduce turbidity, increasing aquatic photosynthesis; allows weak swimmers (e.g., zooplankton) to move upstream
<i>2. Relative width of the interface between realms (e.g., terrestrial-aquatic; river-groundwater)?</i>			
Narrow	Common in this zone and underpins tight hydrological coupling between precipitation and streamflow, leading to steep rising limb of flood hydrograph and rapid rises in water level after heavy rain	Rare in this zone	Rare laterally but common where fine sediments limit hyporheic zone and exchange of river and groundwater, promoting reduced conditions (with biogeochemical implications) in shallow groundwater and soil water

Continued

Table 2.3.2 Preliminary conceptual framework for hypothesizing effects of intermittence on hydrological connectivity and geomorphological, biogeochemical, and ecological processes in different geomorphological zones in IRES—cont'd

Criterion	Hypothesized differences in processes and effects of intermittence in different geomorphological zones in IRES (examples)		
	Upland	Piedmont	Lowland/Floodout
Broad	Rare in this zone	Well-developed hyporheic zone creates broad interface between river and groundwater with long subsurface flowpaths and associated trends in biogeochemistry influencing nutrient concentrations in upwelling water	Extensive floodplains with long lateral hydrological connections that span the broad aquatic-terrestrial interface with implications for fish movement and boom-bust cycles of productivity
<i>3. Connectivity along hydrological dimensions constrained or diffuse?</i>			
Constrained	Well-defined longitudinal hydrological connections mediate transport of sediments and organic materials	Subsurface flowpaths in hyporheic zone mediate transport of dissolved materials	Rare in this zone (except in downwelling termini of some IRES)
Diffuse	Invertebrate drift and fish movement along longitudinal hydrological connections; variation in pool persistence may influence terrestrial oviposition by flying aquatic insects	Across lateral, vertical, and longitudinal hydrological dimensions, active movements of biota may be highly variable and unpredictable	Variation in movement of fish and other aquatic biota (including waterbirds) across inundated floodplains, influencing functional connectivity of metapopulations

materials are governed by drivers associated with standing waters (e.g., stratification, [Boulton et al., 2014](#)). Spatially, the implications of this criterion vary among geomorphological zones ([Table 2.3.2](#)). For example, in the upland zone, flow-mediated processes along longitudinal and lateral hydrological dimensions erode sediment and transport leaf litter from vegetated upper reaches to deposit them in piedmont and lowland/floodout zones. Hyporheic exchange in the piedmont zone relies heavily on advection ([Malard et al., 2002](#)), generating subsurface flowpaths and biogeochemical gradients in all three dimensions simultaneously. Temporally, the criterion is especially relevant to IRES because it explicitly acknowledges that even when flow ceases, hydrological connectivity can persist. Again, this temporal component varies among the geomorphological zones. For example, hydrological connectivity in high-gradient upland zones with limited subsurface flows is likely to be more prone to fragmentation by drying than low-gradient piedmont or lowland zones with substantial groundwater inputs. However, in long pools or stagnant floodwaters of lower gradient zones, the lack of surface flow may allow weak swimmers to move upstream or laterally out on the floodplain ([Sheldon et al., 2010](#); [Table 2.3.2](#)).

The second criterion is drawn from ecological connectivity theory, particularly the “taxonomy” proposed by [Beger et al. \(2010\)](#). One of their couplets distinguishes connectivity across narrow versus broad interfaces where interfaces are defined as entities where elements of two or more realms (e.g., aquatic versus terrestrial) mix. Narrow interfaces are where realms adjoin with relatively little spatial

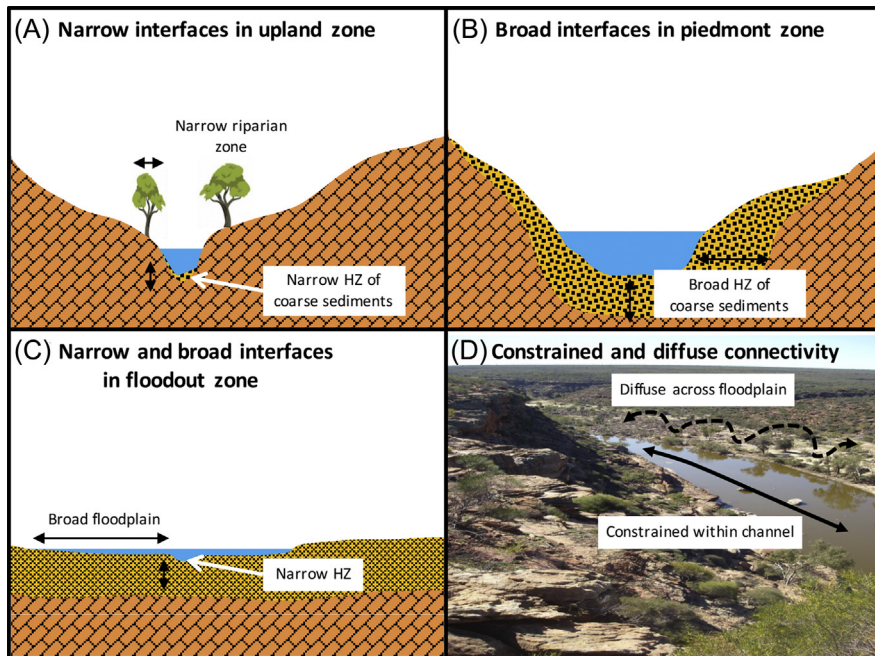


FIG. 2.3.13

Examples in IRES of narrow (A) and broad (B) interfaces (*double-headed black arrows*) and both occurring simultaneously (C), and of constrained and diffuse connectivity (*double-headed black arrows* indicate movement can be in either direction) (D).

separation (Fig. 2.3.13A), such as along riverbanks in the upland zone, whereas broad interfaces have a wide region of mixing between realms (Fig. 2.3.13B), such as the hyporheic zone in the piedmont zone or floodplains in the lowland floodout zone. In IRES, hydrological connectivity across narrow interfaces is likely to be over short distances and transient whereas across broad interfaces, hydrological connectivity will be longer and more persistent.

This criterion's implications relate to distance and water residence time. For example, in upland zones of IRES, tight hydrological coupling between precipitation and streamflow arises from short lateral hydrological connections across narrow riparian zones and steep banks, accelerating rates of water level rise that might drown terrestrial invertebrates (Table 2.3.2; Chapter 4.4). Conversely, the duration and magnitude of “boom” periods of plants, fish, and aquatic invertebrates occupying the broad interfaces represented by floodplains of IRES (Chapter 4.3) rely heavily on the long hydrological connections extending laterally and persisting for months to years after high flows. Narrow and broad interfaces can cooccur in the same river segment, usually in different dimensions (Fig. 2.3.13C), which is why this criterion must be explicitly acknowledged when exploring potential interactions among the three hydrological dimensions.

The third criterion, constrained versus diffuse connectivity (Beger et al., 2010), distinguishes structural and functional connectivity. Throughout this chapter, we have discussed structural connectivity, a

physical attribute of the landscape based on features such as the configuration of patches and their linkages (LaPoint et al., 2015). In contrast, functional connectivity is organism oriented where biological (e.g., shared genetic composition, Chapter 4.10) or ecological features are used to determine whether patches in the landscape are perceived as connected by an organism (Pe'er et al., 2011). In IRES, structural hydrological connectivity may influence functional connectivity even if water itself is not a physical vector (e.g., through modifying movement patterns of flying aquatic insects in catchments of IRES (Chapter 4.3)).

In constrained connections, the link is well defined and spatially stable or subject to only minor movements (at least over timescales of months or years) whereas diffuse connections vary widely in space and time and are difficult to delineate (Fig. 2.3.13D). Often, these diffuse connections are mediated by the movement of biota such as fish or active invertebrates (Gallardo et al., 2009; Kerezy et al., 2013) that create connections between, for example, feeding and breeding areas via routes that are variable, not necessarily water mediated and not always predictable. These connections in IRES are relevant even where these routes leave the water, such as when aquatic insects deposit their eggs terrestrially yet near pools that persist after surface flow has ceased (Table 2.3.2). Again, the two types of connectivity can coincide in the same river segment (Fig. 2.3.13D) and can alternate in response to changing flow regimes.

Although there may be other criteria to add to this framework, the three described here are not mutually exclusive and, in various combinations, characterize most forms of hydrological connectivity in IRES. The criteria by Beger et al. (2010) were proposed as a basis for conservation planning across freshwater, terrestrial, and marine realms, and the framework in Table 2.3.2 could be used similarly when managing the effects of human activities on hydrological connectivity. Many anthropogenic threats to IRES (Chapter 5.1) include disruption of hydrological connectivity (e.g., weirs and dams, extraction of water leading to increased aquatic fragmentation). Another benefit of the framework lies in encouraging researchers to define specific features of hydrological connectivity when describing associations of biogeochemical or ecological processes with aspects of intermittence such as rate of drying or surface flow cessation along one or more hydrological dimensions in IRES.

Finally, the framework can be used to derive hypotheses about how interacting hydrological processes in different dimensions and geomorphological zones along IRES might affect biogeochemical or ecological processes. For example, changes in hydrological connectivity along longitudinal, lateral, and vertical subsurface flowpaths in response to changes in breadth of interface potentially lead to differences in chemical composition of upwelling water in piedmont compared with lowland zones. Other examples are hypothesized in Table 2.3.2. The challenge of unraveling the complexity of these associations, especially in how hydrological connectivity in IRES relates to functional connectivity, is aided by a conceptual framework such as the preliminary one proposed here.

2.3.8 CONCLUSIONS

Hydrological connectivity occurs longitudinally, laterally, and vertically in all flowing waters but is more variable in IRES because of flow intermittence. Spatial and temporal variations in hydrological connectivity are primarily driven by flow regime and catchment geomorphology, often modified by human activities. These variations govern spatial and temporal patterns of surface flow intermittence and, in many cases, drying and rewetting of surface and subsurface water. Concurrent interactions and

discontinuities along all three hydrological dimensions produce complex mosaics of physicochemical patches at different spatial scales whose boundaries fluctuate temporally in response to the pulsed flow regime. Hydrological connectivity in IRES seldom exists as gradual trends in, for example, flows or spatial extent of surface water along the three hydrological dimensions. Instead, they are evident as spatially variable discontinua, according well with current conceptual models of river ecosystems that emphasize nonlinearities in ecological patterns (Poole et al., 2008; Sponseller et al., 2013).

Associations between hydrological connectivity among the three spatial dimensions and the four broad geomorphological zones of IRES (Chapter 2.1) can be explored using three criteria proposed in a preliminary conceptual framework to help generate hypotheses about how hydrological connectivity influences biogeochemical and ecological processes at multiple scales. Researchers are now taking advantage of emerging tools to better quantify spatial and temporal patterns along all three hydrological dimensions at increasingly finer resolution over landscape-level spatial scales. These tools include individual or combined application sensors (e.g., electrical resistance, temperature, camera) along with isotope and other environmental tracers (Costigan et al., 2016) and remote sensing data (Sando and Blasch, 2015) and will substantially improve our understanding of the timing, frequency, magnitude, and rate of exchange along these dimensions. In doing so, we will gain a clearer understanding of how the dynamics of hydrological connectivity in IRES interact with intermittence to influence the biogeochemical and ecological patterns and processes described in the next two sections of this book.

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